AIM:

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Code:-

```
  GNU nano 8.7                                              Leastrecently.c
#include <stdio.h>

int main()
{
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int n = 12;
    int frames[3];
    int time[3];
    int i,j,pos,min;

    int faults = 0, hits = 0, counter = 0;

    for(i=0;i<3;i++)
        frames[i] = -1;

    for(i=0;i<n;i++)
    {
        int found = 0;

        for(j=0;j<3;j++)
        {
            if(frames[j] == pages[i])
            {
                hits++;
                counter++;
                time[j] = counter;
                found = 1;
                break;
            }
        }

^G Help         ^O Write Out    ^F Where Is     ^K Cut          ^T Execute
^X Exit         ^R Read File    ^\ Replace      ^U Paste        ^J Justify
```

```
GNU nano 8.7                                    Leastrecently.c
        }

        if(found == 0)
        {
            min = 0;
            for(j=1;j<3;j++)
            {
                if(time[j] < time[min])
                    min = j;
            }

            frames[min] = pages[i];
            counter++;
            time[min] = counter;
            faults++;
        }
    }

    printf("Total Page Requests = %d\n", n);
    printf("Page Hits = %d\n", hits);
    printf("Page Faults = %d\n", faults);

    float hit_per = (hits*100.0)/n;
    float fault_per = (faults*100.0)/n;

    printf("Hit Percentage = %.2f%%\n", hit_per);
    printf("Fault Percentage = %.2f%%\n", fault_per);

    return 0;
}

^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute     ^C L
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ G
```

OUTPUT:-

```
Nandini Kasare@LAPTOP-NIHKGM40 MINGW64 ~
$ nano Leastrecently.c

Nandini Kasare@LAPTOP-NIHKGM40 MINGW64 ~
$ gcc  Leastrecently.c -o  Leastrecently

Nandini Kasare@LAPTOP-NIHKGM40 MINGW64 ~
$ ./Leastrecently
Total Page Requests = 12
Page Hits = 0
Page Faults = 12
Hit Percentage = 0.00%
Fault Percentage = 100.00%

Nandini Kasare@LAPTOP-NIHKGM40 MINGW64 ~
$ |
```